

Information Technology University of the Punjab
SE460T Cloud Computing with Big Data Applications – Fall 2025
Quiz 2 Section A – November 5, 2025

Name: _____

Roll No.: _____

Time allowed: 30 minutes

Maximum Marks: 25

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1. You create a custom AMI from an EC2 instance and then terminate that instance. What happens to the AMI? [1]
 - a. The AMI is deleted automatically
 - b. The AMI remains available until manually deregistered
 - c. The AMI is moved to S3 Glacier
 - d. The AMI is converted to a snapshot
 2. Which EC2 instance type family is optimized for compute-intensive workloads such as gaming or scientific modeling? [1]
 - a. R family
 - b. T family
 - c. C family
 - d. M family
 3. You have created an EC2 instance with an instance store root volume. What happens to the data on that volume if the instance stops? [1]
 - a. The data is retained
 - b. The data is deleted
 - c. The data is backed up automatically
 - d. The data is moved to EBS
 4. Which Amazon EBS volume type provides the highest performance for I/O-intensive transactional workloads like databases? [1]
 - a. Throughput Optimized HDD (st1)
 - b. Cold HDD (sc1)
 - c. General Purpose SSD (gp3)
 - d. Provisioned IOPS SSD (io2)
 5. An RDS MySQL instance is experiencing high read traffic. Which feature can you enable to scale reads? [1]
 - a. Multi-AZ deployment
 - b. Read replicas
 - c. Cross-region snapshot
 - d. Point-in-time recovery
 6. Which DynamoDB capability automatically deletes expired items based on a timestamp attribute? [1]
 - a. TTL (Time to Live)
 - b. DAX
 - c. Streams
 - d. Global Tables
 7. You want an EC2 instance to automatically install software and configure settings when it launches. Which feature allows you to do this? [1]
 - a. Elastic Load Balancer
 - b. User Data
 - c. Auto Scaling Group
 - d. CloudWatch
 8. The User Data scripts on EC2 instances: [1]
 - a. Run every time the instance is started/stopped.
 - b. Only run once during the first launch of the instance by default.
 - c. Require manual execution after the instance is launched.
 - d. Are only used for Windows instances.
 9. Which of the following is a benefit of using Amazon RDS Multi-AZ deployment? [1]
 - a. Automatic scaling of read workloads
 - b. Automatic failover to a standby instance in case of AZ failure
 - c. Ability to run queries directly on the standby instance
 - d. Lower storage costs compared to single-AZ
 10. Which RDS deployment allows you to have a primary DB instance in one Availability Zone and a synchronous standby in another AZ for high availability? [1]
 - a. Single-AZ deployment
 - b. Multi-AZ deployment
 - c. Read replica
 - d. Aurora Serverless
 11. You need an RDS database that automatically scales storage as your data grows. Which feature should you enable? [1]
 - a. Provisioned IOPS
 - b. Storage Auto Scaling
 - c. Read replica
 - d. Multi-AZ deployment
 12. A company needs to launch a Linux instance that will host a database. The instance needs to stop on occasion to perform maintenance tasks. Which storage solution should they choose for the data? [1]
 - a. Amazon Elastic File System (Amazon EFS)
 - b. Amazon Elastic Block Store (Amazon EBS)
 - c. Amazon FSx for Windows File Server
 - d. Amazon instance store
 13. In DynamoDB, read capacity units (RCUs) and write capacity units (WCUs) are provisioned primarily to: [1]
 - a. Control how much data can be stored in the table and indexes
 - b. Determine how many simultaneous read and write requests the table can handle per second

- c. Specify how frequently the table will be backed up automatically
 - d. Limit the number of secondary indexes that can be added to the table
14. A DynamoDB table has a primary key: UserID (partition key) and OrderDate (sort key). You want to:
 Query all orders of a user sorted by OrderStatus.
 Query all orders of a specific ProductID across all users.
 Which indexes should you create? [1]
- a. LSI on OrderStatus and GSI on ProductID
 - b. LSI on ProductID and GSI on OrderStatus
 - c. GSI on OrderStatus and GSI on ProductID
15. Which of the following best describes an Amazon Machine Image (AMI)? [1]
- a. A template that defines an instance's region, VPC, and subnet
 - b. A template that defines the software configuration, OS, and application stack for an EC2 instance
 - c. A container image for running Docker workloads on EC2
 - d. A snapshot of an S3 bucket configuration
1. In the RDS Challenge Lab, what are the key differences between the architecture at the start of the lab and the architecture after completion of the lab? [3]

The architecture at the start of the lab has MariaDB engine running on the same EC2 instance that is running the web server. The architecture at the end of the lab has the same MariaDB database lying on an RDS for MariaDB instance. The web application in the EC2 instance now accesses the duplicate Database on the RDS instance.

2. After the database has been migrated to RDS for MariaDB, which steps that you perform ensure that the CafeApplication accesses the migrated database and not the local database? [3]

The CaféApplication accesses the migrated database by using the updated secrets in the Secrets Manager, namely, the RDS endpoint, new username, and new password.

3. After duplicating the AMI and launching it on an EC2 instance in the Oregon region, do we make any changes to the new AMI? How do we change the secrets manager parameters in the new Oregon region? [4]

No, we do not make any changes to the new AMI. We change the setapp_parameter.sh file in the previous EC2 instance with updates about region, EC2 URL, etc. This script is then run which configures the Secrets Manager secrets in the region of our new EC2 instance.